

confidentiality requirements.

Passive technology overcomes the limitations of lifetime, reliability and size of implant devices that require a battery. Electromagnetic field received from the interrogator is the source for supplying power to the passive transponder.

Two techniques used for wirelessly supplying the required power to read data from the passive RFID chip are inductive coupling and electromagnetic coupling. In inductive coupling, a time-harmonic magnetic field is generated by the low-frequency current in the interrogator coil. In the near-field region, inductive coupling may occur between this coil and the transponder coil.

Electromagnetic coupling uses electromagnetic waves that propagate from the antenna in the far-field region to power the passive implant chip.

Frequency and read range. An ultra-high-frequency (UHF) device offers significantly higher bandwidth than a low-frequency (LF) or HF device, allowing high data rates and incorporation of privacy/integrity-preserving cryptographic protocols to protect information. Using UHF electromagnetic waves, a high level of miniaturisation of the chip (including antenna) can be achieved.

However, UHF waves may be problematic for human implants as UHF and microwave fields are strongly attenuated by water (which is the primary constituent of human tissue). Consequently, a passive solution allows a very limited communication range (typically shorter than 1m). UHF field transmitted by an interrogator located outside the human body is subject to multiple reflections by environmental objects creating interferences and undesirable shadow zones in which the implanted chip becomes non-detectable by the interrogator.

HFs (in particular, 2.45GHz) may be unsafe for humans for long-term exposure. The resulting range for wireless communication with a UHF implant may then be greatly dependent on the implant's location in the human body. Inductive coupling operating in HF (13.56MHz and below) is one of the most common methods to wirelessly send power and data from

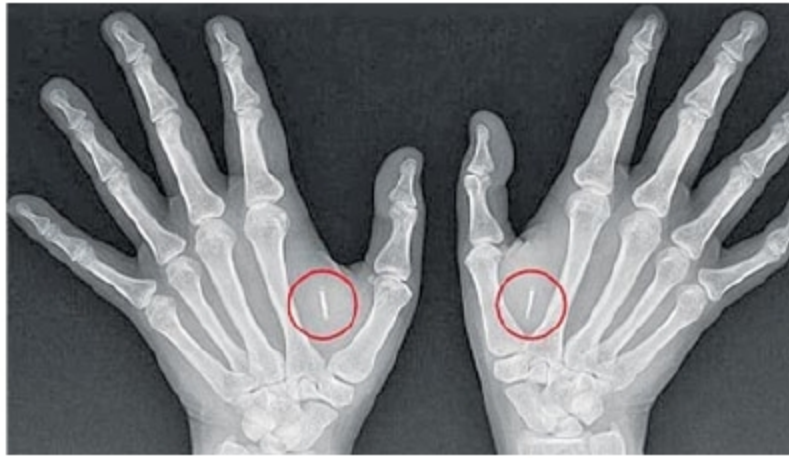


Fig. 7: X-ray showing microchips implanted in both hands to unlock doors without keys (Credit: www.dailymail.co.uk)



Fig. 8: An MRI-compatible pacemaker (Credit: www.medgadget.com)

an off-body interrogator to a device implanted inside the human body.

Biomedical/therapeutic applications of human implant devices. For biomedical applications, human implant devices transmit various chemical or biological parameters measured inside the body. For example, when used on diabetic patients, a wireless alarm unit is implanted in the abdominal region of the patients to continually read the data of a sensor chip sensitive to glucose concentration of surrounding fluids and transmit the data when required.

For therapeutic applications, human implant devices monitor biological activities or physiological functions of the human body and modify them.

Benefits of using microchip implants

The microchip implant can become a very useful tool, especially in emergency situations where instant access to the right medical information can mean the difference between life and death. Here are the various advan-

tages of using implants.

Easy accessibility to medical history. Implanted microchips are useful for patients, particularly those suffering from diabetes, cardiovascular disease or Alzheimer's disease.

The implanted microchip can be used in medical emergencies, especially when a patient is unconscious, to quickly gain access to his/her medical history, such as antibiotics taken in the past, allergies, medication regularly taken and other information that is relevant. The microchip itself need not contain the patient's entire medical history, but rather a unique code that can be used to access the information from a database.

Easy identification. Passports, driver's licences and so on, which are often used for identification purposes, are increasingly containing microchips. For example, passengers can be identified at an airport by scanning the microchip in their passports. It would require minimal changes in infrastructures at airports, and train and bus stations, to transit from scanned passports to scanned implanted chips. People would be identified easily just by walking past a reader.

Institutional memberships and access control. Implanted microchips are convenient and practical to use in workplaces, hotels, gyms and anywhere else where identification is needed to grant access and make payments.

Baja Beach Clubs in Rotterdam, Netherlands and Spain use implanted microchips for members, to avoid long waiting time in lines and for easy access to membership features and facilities. Members can keep track of what they have ordered. The computer automatically debits their bank account by scanning the chips, thus making the process of making payment convenient and easy.

Easy tracking of persons. This is useful for keeping tabs on persons such as patients, children and criminals. For example, it is possible for babies to get mixed up at hospitals,